

7.8 Organic Synthesis (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.8 Organic Synthesis (A Level Only)
Difficulty	Hard

Time allowed: 60

Score: /42

Percentage: /100

Question 1a

Benzocaine is an ester based compound that is used medicinally as a local anaesthetic in pain relievers and cough drops. One of the first industrial processes to manufacture benzocaine was the five-step reaction shown in Fig. 1.1.

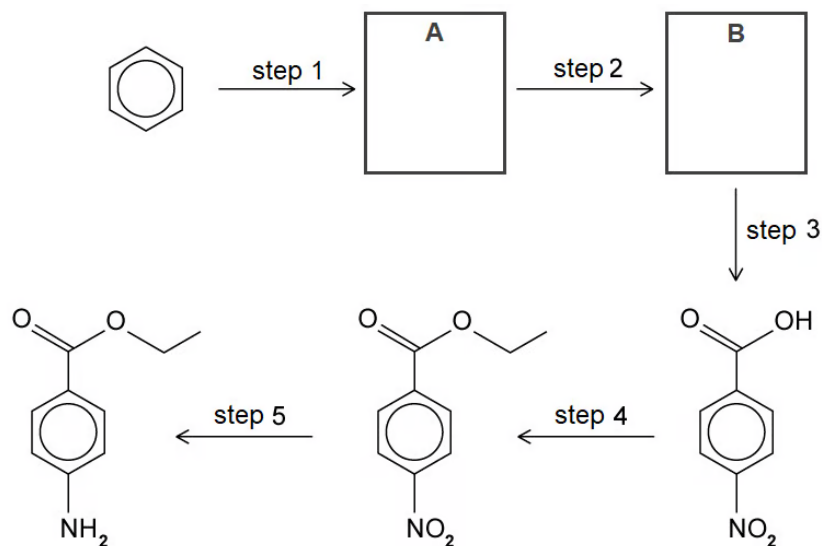


Fig. 1.1

Give the systematic name of benzocaine.

[1 mark]

Question 1b

Step 1 involves the reaction of benzene with bromomethane.

- State a suitable catalyst for step 1.
- Draw a mechanism for step 1. Include all necessary curly arrows, charges and the structure of **A**.

[1]

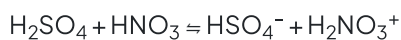
[3]

[4 marks]

Question 1c

Concentrated sulphuric acid and concentrated nitric acid are used to form the ion required for step 2.

The following equilibrium is the first step in the formation of the ion:



i)

State the roles of each acid in this reaction.

[2]

ii)

Use your answer to part (i) to suggest the relative strengths of the acids.

[1]

iii)

Different isomers with the molecular formula $\text{C}_7\text{H}_7\text{NO}_2$ can be formed during step 2.

Draw the structure of the isomer **B**. Explain your answer.

[2]

[5 marks]

Question 1d

Suggest reagents and conditions for steps 3 – 5 of the synthesis.

step 3

step 4

step 5

[4 marks]

Question 1e

Fig. 1.2 shows an alternative reaction scheme for the production of benzocaine.

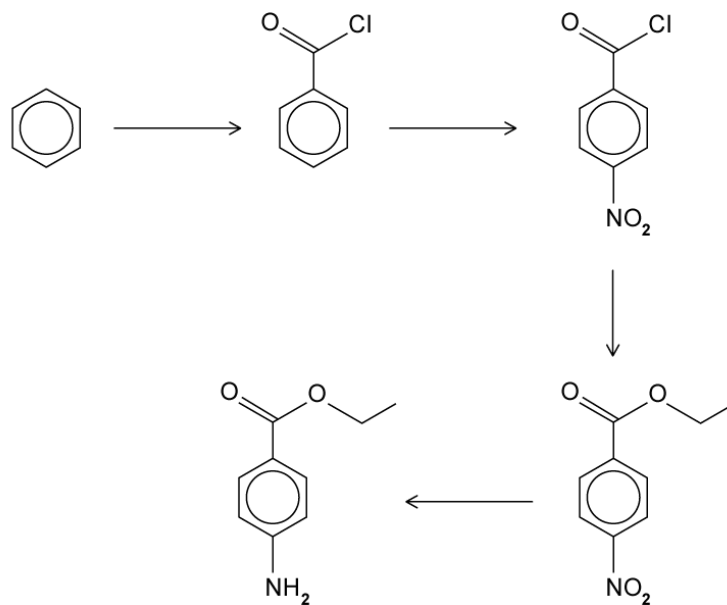


Fig. 1.2

- i)
Suggest why this reaction scheme is an improvement.

[1]

- ii)
Suggest why this reaction scheme may not work.

[1]

[2 marks]

Question 2a

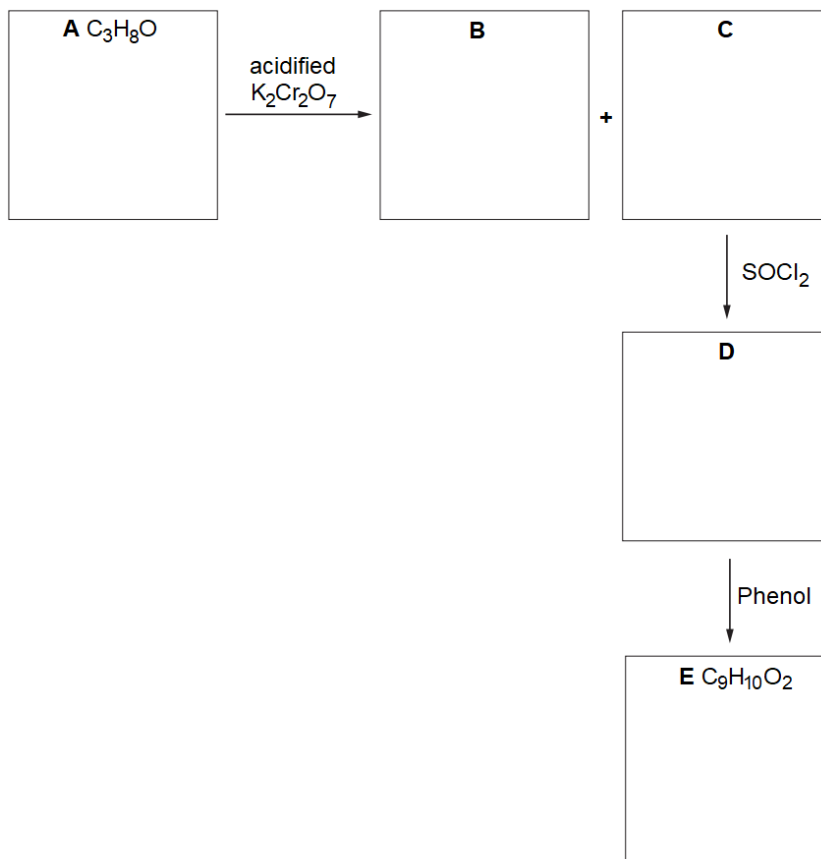
The carbon-13 (^{13}C) NMR spectrum of compound **A**, $\text{C}_3\text{H}_8\text{O}$, contains three peaks.

Compound **A** reacts with acidified potassium dichromate to give a mixture of compounds **B** and **C**.

Compound **C** reacts with sulfur dichloride oxide, SOCl_2 , to form compound **D**.

Compound **D** reacts with phenol to form compound **E**, $\text{C}_9\text{H}_{10}\text{O}_2$.

Draw the structures of compounds **A**, **B**, **C**, **D** and **E** in the boxes.



[5 marks]

Question 2b

A mixture of compound **B** and compound **C** is formed when compound **A** reacts with acidified potassium dichromate.

i)

Explain how the mixture can be separated into compounds **B** and **C**.

[2]

ii)

Suggest an improvement to the reaction with acidified potassium dichromate to ensure the production of compound **C** only.

[1]

[3 marks]

Question 2c

When heated in the presence of sodium hydroxide, compounds **C** and **D** can both react with phenol to form compound **E**.

i)

Draw a mechanism for the formation of compound **E** from compound **D** and an appropriate nucleophile. Include all necessary curly arrows and charges.

[3]

ii)

Under these conditions, the reaction of compound **D** with phenol is faster than compound **C**.

Suggest why the reaction with compound **D** is faster. Explain your answer.

[1]

[4 marks]

Question 3a

Propanone can be synthesised from 2-bromopropane according to the reaction scheme shown in Fig. 3.1.

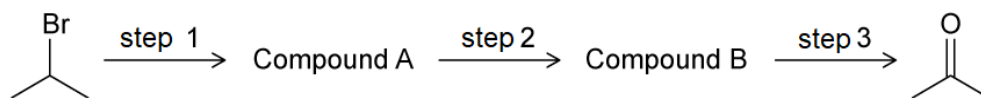


Fig. 3.1

Step 1 is not completed in acidic conditions.

Suggest reagents and conditions for each of steps 1 to 3.

step 1

step 2

step 3

[5 marks]

Question 3b

Suggest **one** reaction, including reagents and conditions, to replace steps 1 and 2.

[2 marks]

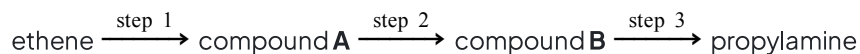
Question 3c

Using structural formulae, write an equation for the reaction of compound **B** with an oxidising agent, $[O]$, to form propanone.

[1 mark]

Question 4a

Ethene is used in the radical-free synthesis of 1-aminopropane as shown.



Identify which step of the reaction scheme could use hydrogen cyanide. Explain your answer.

[2 marks]

Question 4b

Draw the **skeletal** structure of compound **A** and state the reagents required for its formation.

[2 marks]

Question 4c

The final step of the reaction scheme forms 1-aminopropane from compound **B**.

Name compound **B** and identify the type of reaction that compound **B** undergoes to form propylamine. Explain your answer.

[2 marks]

Model Answers: Hard

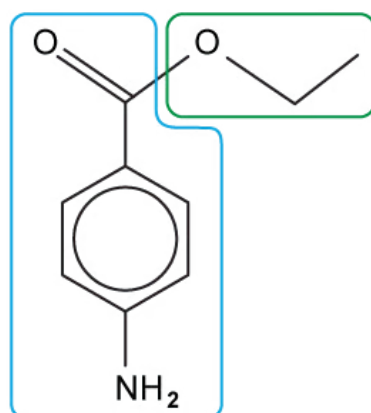
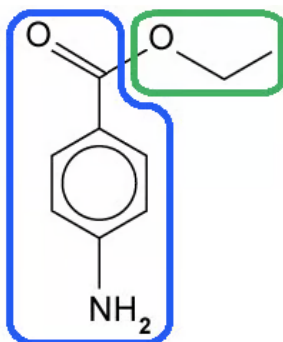
1a

a) The systematic name of benzocaine is:

- Ethyl 4-aminobenzoate; [1 mark]

[Total: 1 mark]

- The question tells you that benzocaine is an ester
 - This means that the name will follow the general alkyl alkanoate format for naming esters
- Remember:** The direct formation of an ester involves the reaction of an alcohol and a carboxylic acid
 - Alcohol + acid $\rightarrow$ Ester



- The alcohol (green) used gives the alkyl / first portion of the name
 - The alcohol contains two carbons and is, therefore, ethanol

- The carboxylic acid (blue) used gives the alkanoate / second portion of the name
 - The carboxylic acid is benzoic acid with an amine group in the 4-position on the benzene ring
 - Therefore, the acid is 4-aminobenzoic acid
- So, to form benzocaine directly from an alcohol and a carboxylic acid:
 - Ethanol + 4-aminobenzoic acid $\rightarrow$ Ethyl 4-aminobenzoate + water

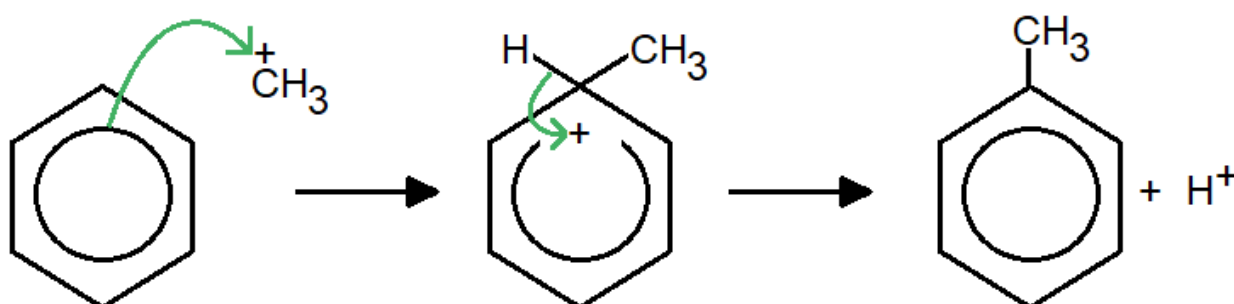
1b

b)

i) A suitable catalyst for step 1 is:

- AlBr_3 / aluminium tribromide
OR
 FeBr_3 / iron tribromide; [1 mark]

ii) The mechanism for step 1, including all necessary curly arrows and charges, is:



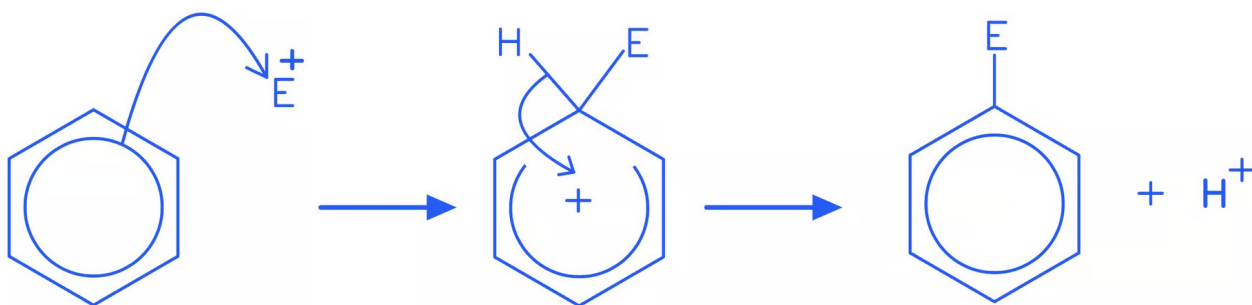
- Curly arrow from the ring inside benzene to the + carbon / C^+ ; [1 mark]
- Correct intermediate structure with horseshoe covering more than half of the benzene but not extending past carbon 2 or 6; [1 mark]
- Curly arrow from the C-H bond back to the + within the ring

ANDCorrect structure of **A** / methylbenzene; [1 mark]**[Total: 4 marks]**

- The reaction of benzene with bromomethane is a Friedel-Crafts alkylation reaction
 - **Careful:** This is most commonly performed with a chloroalkane and an AlCl_3

catalyst

- But the question tells you that one of the reactants is bromomethane, which means that an AlBr_3 or FeBr_3 is required
- The catalyst is required to form the $^+\text{CH}_3$ electrophile for this reaction
 - $\text{CH}_3\text{Br} + \text{AlBr}_3 \rightarrow ^+\text{CH}_3 + \text{AlBr}_4^-$
- The mechanism then continues as a standard electrophilic substitution reaction



- The electrophile bonds to the benzene ring
- This forms an intermediate with a partially delocalised electron system which is shown as a horseshoe with + inside
- The intermediate loses a proton, restoring the delocalised electron system and regenerating the catalyst
- There are extra marking points for electrophilic substitution mechanisms:
 - The + must be on the correct carbon in the $^+\text{CH}_3$ electrophile
 - The first arrow must start touching the ring, not start inside the ring
 - The horseshoe in the ring must be centred about carbon 1 and not extend past carbon 2 or 6

1c

c)

i) The roles of each acid are:

- Sulfuric acid is acting as an (Brønsted–Lowry) acid
AND
Because it is donating a proton / hydrogen ion / H^+ ; [1 mark]
- Nitric acid is acting as a (Brønsted–Lowry) base
AND
Because it is accepting a proton / hydrogen ion / H^+ ; [1 mark]

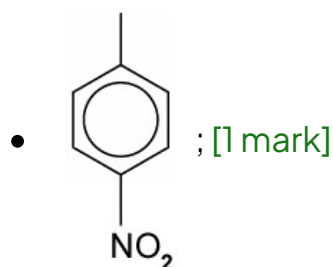
ii) Comparing the strengths of the acids:

- Sulfuric acid is stronger than nitric acid (because it loses / donates a proton)

OR

Nitric acid is weaker than sulfuric acid (because it gains / accepts a proton); [1 mark]

iii) The structure of isomer **B** is:



- The methyl / CH₃ group on methylbenzene (is electron donating which) activates the 2, 4 and 6 positions on the benzene ring

AND

The nitro / NO₂ group must attach / bond to carbon-4 (to form the product of step 3); [1 mark]

[Total: 5 marks]

- The reversible arrow in this reaction is irrelevant
- The reaction to form the nitronium ion occurs in a number of steps but the overall equation for the formation of the nitronium ion is
 - $\text{HNO}_3 + 2\text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + 2\text{HSO}_4^- + \text{H}_3\text{O}^+$
- For part (i)
 - You are basically being asked what the roles of the acids are
 - The sulfuric acid is losing / donating a proton
 - This means that the sulfuric acid is acting as a Brønsted–Lowry acid
 - The nitric acid is gaining / accepting a proton
 - This means that the nitric acid is acting as a Brønsted–Lowry base
 - One of the first definitions of an acid is about being a proton donor – this is the definition
- For part (ii)
 - Both acids would normally lose / donate a proton
 - Therefore, sulfuric acid must be a stronger acid as it is losing / donating a proton to nitric acid
 - Careful:** Since you are being asked to suggest the relative strengths,

your answer must include comparative language e.g. sulfuric acid is **stronger** than nitric acid

- For part (iii)
 - The methyl / CH_3 group of methylbenzene donates electron density to the π -system of the benzene ring
 - This activates the 2, 4 and 6 positions for possible attack
 - Looking at the product of step 3, the nitro / NO_2 group must have reacted at carbon-4

1d

d) The reagents and conditions for steps 4 and 5 of the synthesis are:

Step 3

- Reflux with potassium manganate(VII) / KMnO_4 / MnO_4^-
AND
Reflux; [1 mark]

Step 4

- Warm with ethanol; [1 mark]
- Concentrated sulfuric acid catalyst; [1 mark]

Step 5

Alternative reagents 1

- Tin and concentrated hydrochloric acid **OR** Sn / HCl
AND
Under reflux; [1 mark]

Alternative reagents 2

- Iron and concentrated hydrochloric acid **OR** Fe / HCl
AND
Under reflux; [1 mark]

Alternative reagents 3

- Nickel and hydrogen

OR

Ni / H₂; [1 mark]

[Total: 4 marks]

- Step **3** is the complete oxidation of an alkyl arene
 - Alkyl side-chains in alkyl arenes will oxidise to carboxylic acids when refluxed with alkaline potassium manganate(VII)
 - They are then acidified with dilute sulfuric acid (H₂SO₄) but this is not required for the mark
- Step **4** is an esterification reaction
 - This means that you need the corresponding alcohol to form the ester
 - From the reaction scheme, you can see that the alcohol must have an ethane-based carbon chain
 - Esterification also requires warming with a concentrated sulfuric acid catalyst
- Step **5** is the reduction of a nitro compound to its corresponding amine
 - This is done using Sn / HCl or Fe / HCl or Ni / H₂
 - **Careful:** LiAlH₄ and NaBH₄ are not suitable for this reduction reaction

1e

OR

Ni / H₂; [1 mark]

[Total: 4 marks]

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 - This is done using Sn / HCl or Fe / HCl or Ni / H₂
 - **Careful:** LiAlH₄ and NaBH₄ are not suitable for this reduction reaction

1e

e)

i) This reaction scheme is an improvement because:

- There are fewer / less steps (reaction scheme); [1 mark]

ii) This reaction scheme may not work because:

- The nitration step / nitration of benzoyl chloride may / does not happen at carbon 4 of the benzene ring; [1 mark]

[Total: 2 marks]

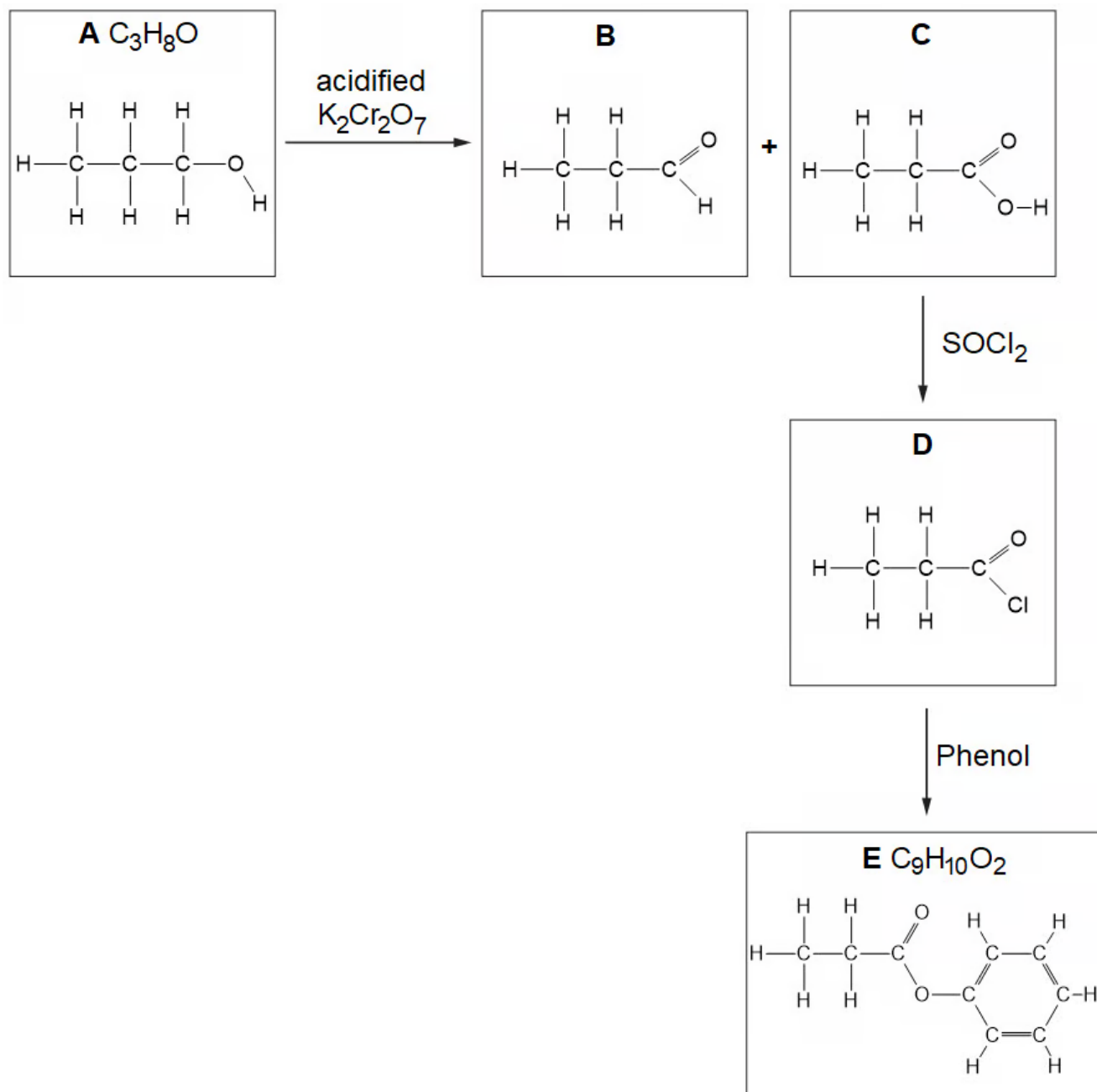
- Chemists aim to have fewer steps in any reaction scheme
 - This typically improves efficiency / overall atom economy
- The key changes in Fig. 1.2 are the first two steps with the Friedel Crafts acylation of benzene and the nitration of the resulting benzoyl chloride
- Fig. 1.1 has the extra step of converting the 4-nitromethylbenzene into 4-

nitrobenzoic acid - you are not expected to know the reagents, conditions or mechanism for this step

- The final two esterification and reduction steps of both reaction schemes are the same
 - Therefore, the difference **must** be in those first two steps
- The first step of both reaction schemes might use different chemicals to form different products but it is ultimately an electrophilic substitution reaction
- In the second step in Fig. 1.1, nitrating the methylbenzene will produce 4-nitromethylbenzene
- In the second step in Fig. 1.2, nitrating the benzoyl chloride does **not** preferentially form 4-nitrobenzoyl chloride
 - It forms 3-nitrobenzoyl chloride as the acyl chloride group directs attack to carbon 3

2a

a) The structures of compounds **A**, **B**, **C**, **D** and **E** are:



- Correct structure of compound **A** / propan-1-ol; [1 mark]
- Correct structure of compound **B** / propanal; [1 mark]
- Correct structure of compound **C** / propanoic acid; [1 mark]
- Correct structure of compound **D** / propanoyl chloride; [1 mark]
- Correct structure of compound **E** / phenyl propanoate; [1 mark]

[Total: 5 marks]

- The reaction of **A** with acidified potassium dichromate should suggest that the compound could be a primary alcohol, secondary alcohol or aldehyde
 - Only a primary alcohol could react with acidified potassium dichromate to

form two possible products

- **Remember:** Primary alcohol $\rightarrow$ aldehyde $\rightarrow$ carboxylic acid
- The question gives the molecular formula of **A** and that its ^{13}C NMR spectrum contains 3 peaks
 - The NMR data suggests that **A** is a primary alcohol or an ether (which are not on the specification)
 - Therefore, **A** is propan-1-ol / $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- Depending on the conditions, propan-1-ol will react with acidified potassium dichromate to form propanal or propanoic acid
 - **Careful:** Compound **C** has to be propanoic acid / $\text{CH}_3\text{CH}_2\text{COOH}$ for the next step to be able to occur
 - Therefore, **B** is propanal / $\text{CH}_3\text{CH}_2\text{CHO}$
- SOCl_2 is a reagent that is used to convert carboxylic acids into acyl chlorides
 - Therefore, **D** is propanoyl chloride / $\text{CH}_3\text{CH}_2\text{COCl}$
- The subsequent reaction with phenol is an esterification reaction
 - **Careful:** Do not be put off by the ring inside phenol and automatically assume that it is an electrophilic substitution reaction
 - **Remember:** Alcohol + acyl chloride $\rightarrow$ ester + HCl
 - Phenol + propanoyl chloride $\rightarrow$ phenyl propanoate + HCl
 - Therefore, **E** is phenyl propanoate / $\text{CH}_3\text{CH}_2\text{COOC}_6\text{H}_5$
- Make sure that you draw the structures of all the compounds as instructed in the question

2b

b)

i) The mixture can be separated into compounds **B** and **C** by:

- Distillation
- AND**
- (Because,) **C** has a higher boiling point than **B**; [1 mark]
- Compound **C** can form hydrogen bonds which require more energy to overcome; [1 mark]

ii) An improvement to the reaction with acidified potassium dichromate to ensure the production of compound **C** only is to:

- Reflux (**A** with the acidified potassium dichromate); [1 mark]

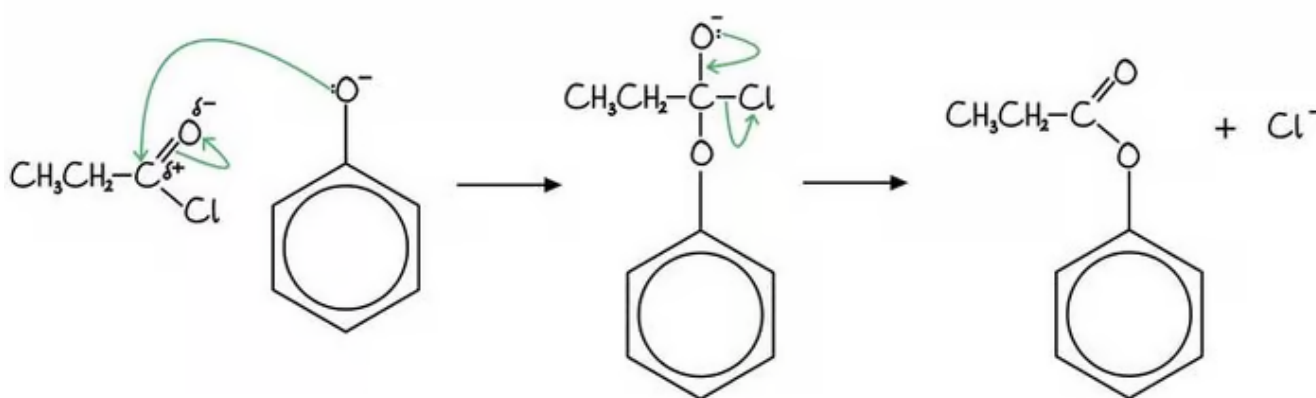
[Total: 3 marks]

- From part (a), compound **B** is propanal and compound **C** is propanoic acid
 - Compound **C** has lone pairs on the oxygen atoms **AND** has a hydrogen atom attached to the electronegative oxygen atom
 - This means that it is able to hydrogen bond
 - Hydrogen bonding is the strongest intermolecular force and requires a large amount of energy to overcome, resulting in high boiling points
 - Compound **B** has a dipole on the carbonyl group
 - But, the energy required to overcome this is much less than the energy required to overcome hydrogen bonding
 - This means that compounds **B** and **C** have different boiling points and can be separated by distillation
 - It is not necessary to specify simple or fractional for the mark
- To ensure that all of the propan-1-ol / **A** is oxidised into propanoic acid / **C**, the reaction with acidified potassium dichromate should be performed under reflux

2c

c)

i) The mechanism for the formation of compound **E** from compound **D** and an appropriate nucleophile is:



- Curly arrow from the lone pair on the oxygen atom of the nucleophile to the δ^+ carbon
AND

Curly arrow from the carbonyl / C=O bond to the oxygen atom in propanoyl chloride; [1 mark]

- Correct intermediate structure; [1 mark]
- Curly arrow from the lone pair on the O⁻ on the intermediate to the C-O bond
AND
Curly arrow from the C-Cl bond to the chlorine atom; [1 mark]

ii) The reaction of compound **D** with phenol is faster because:

- The partial charge on the carbonyl carbon in compound **D** is more positive than compound **C**
AND
Which makes compound **D** more susceptible to attack from the nucleophile / phenoxide ion; [1 mark]

[Total: 4 marks]

- For part (i)
 - The mechanism for this reaction is a nucleophilic addition-elimination reaction
 - **Careful:** The question states that the mechanism should be drawn with an appropriate nucleophile
 - In this case, the nucleophile is the phenoxide ion / C₆H₅O⁻

Curly arrow from the C-Cl bond to the chlorine atom; [1 mark]

ii) The reaction of compound **D** with phenol is faster because:

- The partial charge on the carbonyl carbon in compound **D** is more positive than compound **C**

AND

Which makes compound **D** more susceptible to attack from the nucleophile / phenoxide ion; [1 mark]

[Total: 4 marks]

- For part (i)
 - The mechanism for this reaction is a nucleophilic addition-elimination reaction
 - **Careful:** The question states that the mechanism should be drawn with an appropriate nucleophile
 - In this case, the nucleophile is the phenoxide ion / $\text{C}_6\text{H}_5\text{O}^-$
 - This is formed by sodium hydroxide deprotonating phenol :
 $\text{C}_6\text{H}_5\text{OH} + \text{NaOH} \rightarrow \text{C}_6\text{H}_5\text{O}^- + \text{Na}^+ + \text{H}_2\text{O}$
- For part (ii)
 - Compounds **C** and **D** will both react with the phenoxide ion
 - Therefore, any difference in the speed / rate of reaction must be due to the structure of compounds **C** and **D**
 - Both compounds have a carbonyl group
 - Compound **C** has the carbonyl / $\text{C}=\text{O}$ group attached to an OH group
 - Compound **D** has the carbonyl / $\text{C}=\text{O}$ group attached to a Cl atom
 - Oxygen and chlorine are both electronegative
 - For compound **D** to react faster, chlorine must be more electronegative than oxygen
 - This results in compound **D** have a larger partial positive charge on the carbonyl carbon, which makes it more attractive to a nucleophile

3a

a) The reagents and conditions for each step are:

Step 1

- Ethanolic / alcoholic sodium hydroxide / NaOH
- OR**

Ethanol / alcoholic potassium hydroxide / KOH; [1 mark]

- Heated under reflux; [1 mark]

Step 2

- Water vapour / H_2O (g) / Water at $300\text{ }^\circ\text{C}$; [1 mark]
 - Concentrated sulfuric acid / H_2SO_4 catalyst
- OR**
- Concentrated phosphoric acid / H_3PO_4 catalyst; [1 mark]

Step 3

- Heat / distil with (acidified) potassium dichromate (VI) / $\text{K}_2\text{Cr}_2\text{O}_7$ solution; [1 mark]

[Total: 5 marks]

- You are told that step 1 is not completed in acidic conditions
 - If you complete step 1 in alkaline / basic conditions, then you can:
 - Form propene via an elimination reaction - this would then allow for two further steps to form propanone
 - Form propan-2-ol via a nucleophilic substitution reaction - this would only require one further step to form propanone
 - Therefore, step 1 is the elimination reaction of 2-bromopropane to form propene
 - This requires refluxing with alcoholic a base like sodium hydroxide or potassium hydroxide
- Step 2 converts propene into propan-2-ol
 - This is a hydration reaction to form the secondary alcohol which requires water with a concentrated H_2SO_4 or H_3PO_4 catalyst at $300\text{ }^\circ\text{C}$ (and 7 MPa)
- Step 3 is the reduction of propan-2-ol into propanone
 - This is an oxidation reaction which requires heating / distilling with acidified potassium dichromate(VI) solution
 - You do not need to specify distillation as a secondary alcohol will oxidise to a ketone with no further oxidation using acidified potassium dichromate

3b

b) **One** reaction to replace steps 1 and 2 is:

- Nucleophilic substitution; [1 mark]

- Nucleophilic substitution, [1 mark]
- Using aqueous sodium hydroxide / NaOH (aq) heated under reflux; [1 mark]

[Total: 2 marks]

- **Careful:** Take your time, this question has the distraction of compound **A** in the middle of where you need to look
- **Tip:** Remove the distraction of the reaction scheme by drawing the 2-bromopropane reactant and the propan-2-ol product
 - You should see that the bromine has been substituted by a hydroxyl group
- **Remember:** Aqueous conditions favour substitution
 - This means that you are using nucleophilic substitution from AS Chemistry with aqueous sodium hydroxide, heated under reflux

3c

c) The equation for the reaction of compound **B** with an oxidising agent, [O], to form propanone is:

- $\text{CH}_3\text{CHOHCH}_3 + [\text{O}] \rightarrow \text{CH}_3\text{COCH}_3 + \text{H}_2\text{O}$; [1 mark]

[Total: 1 mark]

- **Careful:** The question asks for structural formulae
 - Compound **B** is propan-2-ol with the structural formula $\text{CH}_3\text{CHOHCH}_3$
 - Propanone has the structural formula CH_3COCH_3
- Step 3 is an oxidation reaction where the oxidising agent, [O], removes hydrogen from the secondary alcohol to form a ketone
 - The second product is water

4a

a) The step of the reaction scheme that could use hydrogen cyanide is:

Step 2 AND at least one of the following:

- It cannot be step 1 because the cyanide ion / hydrogen cyanide will not react directly with ethene

OR

It cannot be step 1 because the cyanide ion (is a nucleophile and) will not attack the C=C / double bond of ethene due to repulsion; [1 mark]

the C=C / double bond of ethene due to repulsion; [1 mark]

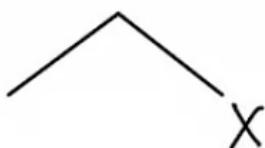
- It cannot be step 3 because the reaction with the cyanide ion / hydrogen cyanide produces a nitrile / does not produce an amine; [1 mark]

[Total: 2 marks]

- You cannot score full marks without identifying the correct step
- Your understanding of command words and exam technique is important in this question
 - There are three steps and two marks which combined with "explain your answer" suggest that you are going to gain marks for explaining why two steps cannot be the reaction with hydrogen cyanide
- As the mark scheme shows, you have to consider how the reactant of each step would behave with hydrogen cyanide or, more specifically, the cyanide ion

4b

b) The skeletal structure of compound **A** is:



- Correct skeletal formula of compound **A** with any appropriate halogen; [1 mark]
- Reagent = corresponding hydrogen halide compound to match the skeletal formula; [1 mark]

[Total: 2 marks]

- Compound **A** must be able to react with the cyanide ion
- Ethene is likely to undergo electrophilic addition
 - This means that you should automatically be thinking H_2 , HX , X_2 , H_2SO_4
 - As the next reaction is likely to be a nucleophilic substitution with CN^- , compound **A** needs to be monosubstituted and contain **one** polar bond
 - Therefore, the only realistic option is the reaction with HX
 - **Remember:** The X stands for a halide ion

4c

Answer for 4c:

1) Compound **B** is:

- Propanenitrile; [1 mark]

The type of reaction that compound **B** undergoes to form propylamine is:

- Reduction

AND

The nitrogen / nitrile group has gained hydrogen; [1 mark]

[Total: 2 marks]

- Examiners often comment about the apparent difficulty that some students have naming compounds
 - Nitriles are an examiner favourite for this because:
 - The carbon chain increases, e.g. eth- to prop-
 - Students forget the letter “e” and sometimes it is marked as essential
 - So, make sure you can name nitriles correctly
- The nitrile group has gained hydrogen as it converted into an amine
 - **Remember:** Look for the GCSE identifiers of oxidation and reduction first because it can save time when identifying types of reaction

